
To design an optimal Auto Workshop facility



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Assessment Rubrics

Knowledge Component	Domain	PLO mapped	Taxonomy Level	Max Marks	Obtained Marks
Each group should use AHP to select the appropriate location of facility (Evaluate)	Cognitive	2	C-4	5	
Apply Factory cad / linear programming to model the problem	Cognitive	1	C-2	5	
Analyze different parameters using Craft method or any other optimization method	Cognitive	2	C-3	5	
Analyze designing/ improvement of an industrial facility using graph based method	Cognitive	2	C-4	5	
Consider all other factors of facility design (Analyze)	Cognitive	2	C-4	5	
Create Design of facility	Cognitive	3	C-6	5	
Apply facility design principles to one's own work, as a member and /or leader in a team	Affective	11	A-2	5	
Manage facility design project in the broader context of innovation and technological developments.	Affective	11	A-2	5	

Abstract

The ultimate aim of the project is to apply engineering planning aspects and algorithms to design a Facility of automobile workshop which has a particular importance in each and every city, so before the development of auto workshop some planning factors must be considered like, need of this facility, public convenience, right location, right amount and investment. The report is about the alternates about facility, the location alternatives, decision making strategies, and design of facility, algorithmic based approach to optimize the design, interior and exterior of workshop. It is important to consider all necessary factors before structuring the design, this can save time and investment for this reason, a design is made in 2D designing software so any changes can be made within the virtual reality before implementing the design. The report is divided into four chapters, first one contains an introduction, next is the related work, and then comes methodology and finally improvements and a conclusion.

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1. Introduction

Industrial facilities means facilities such as, factories, workstations, warehouses, fuel stations, service areas are structured in such a way that maximize the flow and optimize the resource utilization. Facilities planning [1] determines how an activity to be performed in certain environment. It is the measure of tangible, fixed resources should contribute to meeting the activity's objectives. The aim of facility planning is to define the problem define the objective of the facility and specify the primary and support activities to meet the overall objectives of facility design [2].

1.1. Problem Statement

The aim of the project is to plan and design the automobile workshop layout, which optimize the overall available resource utilization and maximize the flow of activities [3] inside the workshop.

1.2. Objectives

Following are some of the objectives to accomplish the facility design.

- Select appropriate tools for decision making for location of facility (Cognitive ,PLO-2)
- Create different designs of the layouts (Cognitive, PLO-3)
- Create a feasible design and layout of the required facility using 2D Designing software (Cognitive, PLO-1)
- Consider all parameters that are required for automobile workshop design (PLO-1, PLO2, PLO-4)
- The facility must consider important health and safety factors (PLO-6)

1.3. Overview of the project

In this project, different alternatives for the location of the workshop are identified and AHP method of decision making is implemented to find the optimal location of facility, after finalizing the location two different layouts designs are proposed using graph based method and analysis is made on the basis of number of workstations and maximum flow of activities in the shop.

The following are the main parameters and tools used in the project:

-
- Analytical Hierarchy Process (AHP) [4], to decide the optimal location of auto workshop
 - Graph based method [5] to create and optimize the layout of workshop
 - Two different layouts are proposed, depending upon the location and available space, again AHP is employed to find the best layout design
 - Other interior and exterior requirements like lighting, sign board, interior color, safety equipment, material handling [7] area, parking area, and necessary stuff is identified.

2. Literature Review

An automobile repair shop (also known regionally as a garage or a workshop) is an establishment where automobiles are repaired by auto mechanics and technicians.

Also there are online automobile repair shops [8] (mobile mechanics) provide doorstep repair services and home delivery of new and used auto parts of different late model and classic cars whose parts are not widely available in the market.

Online automobile [9], [10] service marketplace Open-bay is providing consumers choice and convenience when it comes to getting their vehicles serviced. Consumers are provided with choice in service centers to select from; service department at a car dealership, independent automotive repair facilities, and many others. Choices also include mobile mechanics such as Repair Smith and Wrench where service is performed at a location you dictate (home, office, other).

In countries such as the UK, the mobile car body repair sectors has experienced high growth by way of mobile SMART Repair companies [8] providing mobile car body repair services, such as Bumper Repairs, auto body repair, paint-less dent repair and paintwork defect repairs to private and commercial consumers, typically within the industry framework of refinishing vehicle damage on a localized basis, where the area of damage being repaired is not in excess of an A4 sheet of paper.

But the importance of local auto workshop cannot be denied. One of the benefits of visiting an auto repair shop regularly, even when your car's issue doesn't seem acute, is that you can ensure that your car is well maintained and running properly. If you ignore an issue with your car, that issue will gradually become exacerbated as you ignore it

Automobile service workshops are facing a lot of problems relating employee retention [11] now a day. As employees are the most precious and important asset of an organization, so, organization

should take measures to encourage employees to stay in the organization for the maximum period of time, or until the completion of the project.

Automotive repair shops that specialize in bodywork repair are known as body shops. They offer paintwork repairs to scratches, scuffs and dents, as well as repairs to the bodies of vehicles damaged by collisions. Many body shops now offer paint-less dent repair and auto glass replacement. Automotive repair shops that specialize in auto glass repair are known as auto glass repair shops. They offer auto glass repairs to chips, cracks and shattered glass. The types of glass they repair include windshields, car windows, and quarter glass and rear windows. This type of damage is often caused by hail, stones, wild animals, fallen trees, automobile theft and vandalism. Automobiles are need for everyone in this world. People use this for transportation purposes and several other reasons like trading of goods and services from one place to another. But the most common reason for the massive use of vehicles is the transportation of passengers from one place to another, similarly the maintenance and their repair has also multiplied, auto workshop on the road side are the best option to provide this service[12]

3. Methodology

In this chapter different alternatives, algorithmic and optimization techniques are discussed in detail. AHP technique [13] is employed for deciding the location of facility, Graph based method is used for optimizing the layout and is employed to create the proposed 2D design of layout.

3.1. Location Identification using AHP (Cognitive, PLO-2)

Three different alternatives are proposed for the design as shown in figure 3.1, these include

1. Location1 Taxila Main
2. Location 2 Brahma Interchange
3. Location3 Hassan Abdal

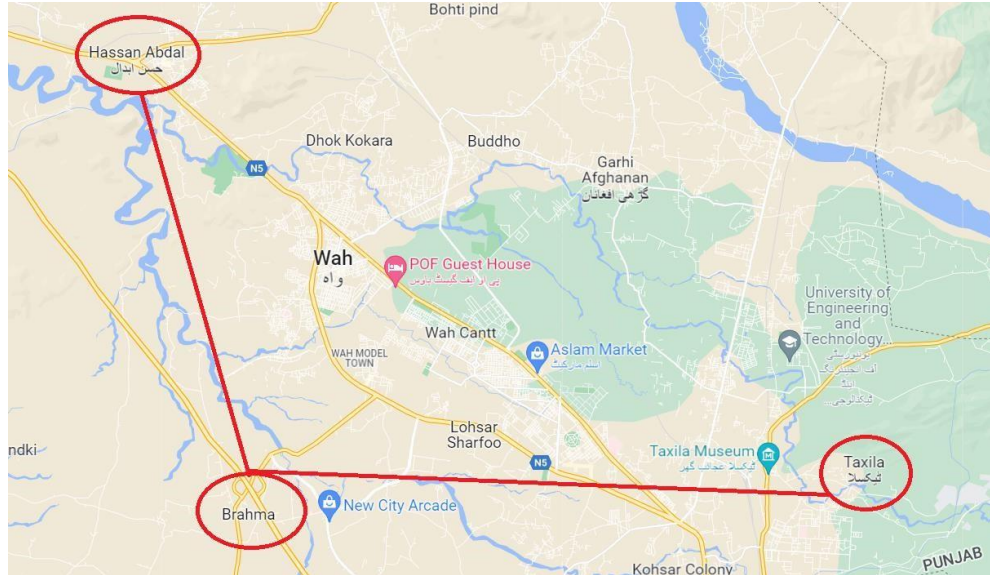


Figure 3.1 Facility location alternatives, picture from google map

3.1.1. Decision making technique

Analytical Hierarchy Process is a tool or technique which is used to evaluate different alternatives for proposing a solution and also for decision making. In our case the online free AHP tool is used to predict the analysis based on the following criteria

- Cost of land
- Need
- Profit
- Nearby competition
- Convenience for Public

Following are the screenshots of software interface, the figure 3.2 provides the overall weightage assigned by us the basis of criteria mentioned above.

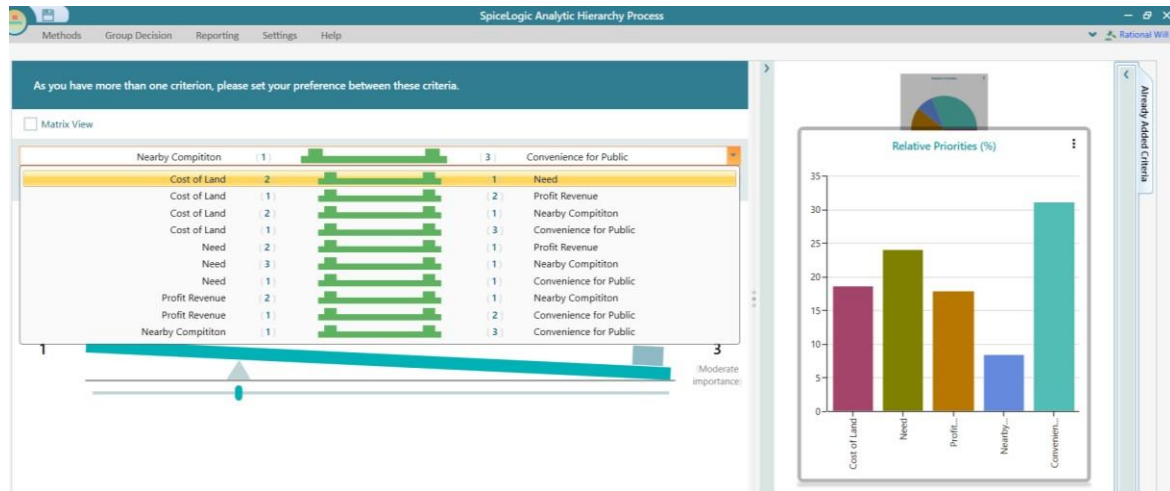


Figure 3.2 Software interface for AHP technique of selecting location alternatives

Software enables us to priorities the criteria, given below in figure 3.3.

Table 3. 1 AHP matrix for five different criterions

	Cost of Land	Need	Profit Revenue	Nearby Compititon	Convenience for Public	Priorities
Cost of Land	1	2	0.5	2	0.333	0.185
Need	0.5	1	2	3	1	0.24
Profit Revenue	2	0.5	1	2	0.5	0.179
Nearby Compititon	0.5	0.333	0.5	1	0.333	0.084
Convenience for Public	3	1	2	3	1	0.311

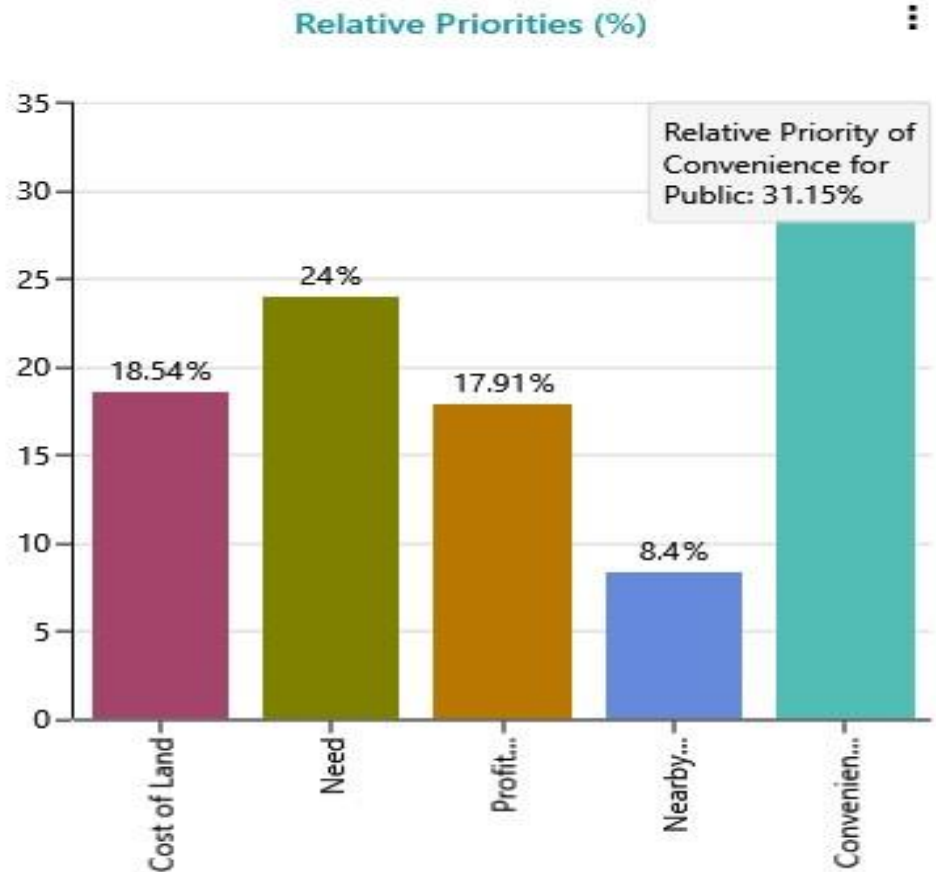


Figure 3.3 Result of AHP criteria in the form of Bar Graph

After analyzing the criterion results it is concluded that, convenience of public is our most important priority, which is given 31% priority, need for public is of secondary importance with 24% weightage and our third priority is cost of land which is given 18% weightage.

Now let's evaluate the location alternatives based on these priorities. Based upon cost of land (shown below in the form of bar graph in figure 3.4) Brahma is the best option with weightage 61%, on criteria need for public. Again, Brahma is the best option with 49% weightage. While on the profit revenue and nearby competition Taxila is the better option among Brahma and Hasan Abdal with 49% weightage in both criteria. On the other hand, on upon convenience for public again Brahma is better option with weightage 49%.

So, based upon all criteria the winning alternative is Brahma, which is proved the best option as shown in figure 3.5, in which cost of land contributes 61%, this means we have a great margin over the cost of land on Brahma Roadside. Public convenience and need contribute 49%, although profit revenue is less at this place which is involved 20% but there is a little competition as compared to other alternate areas like Taxila and Hasan Abdal.

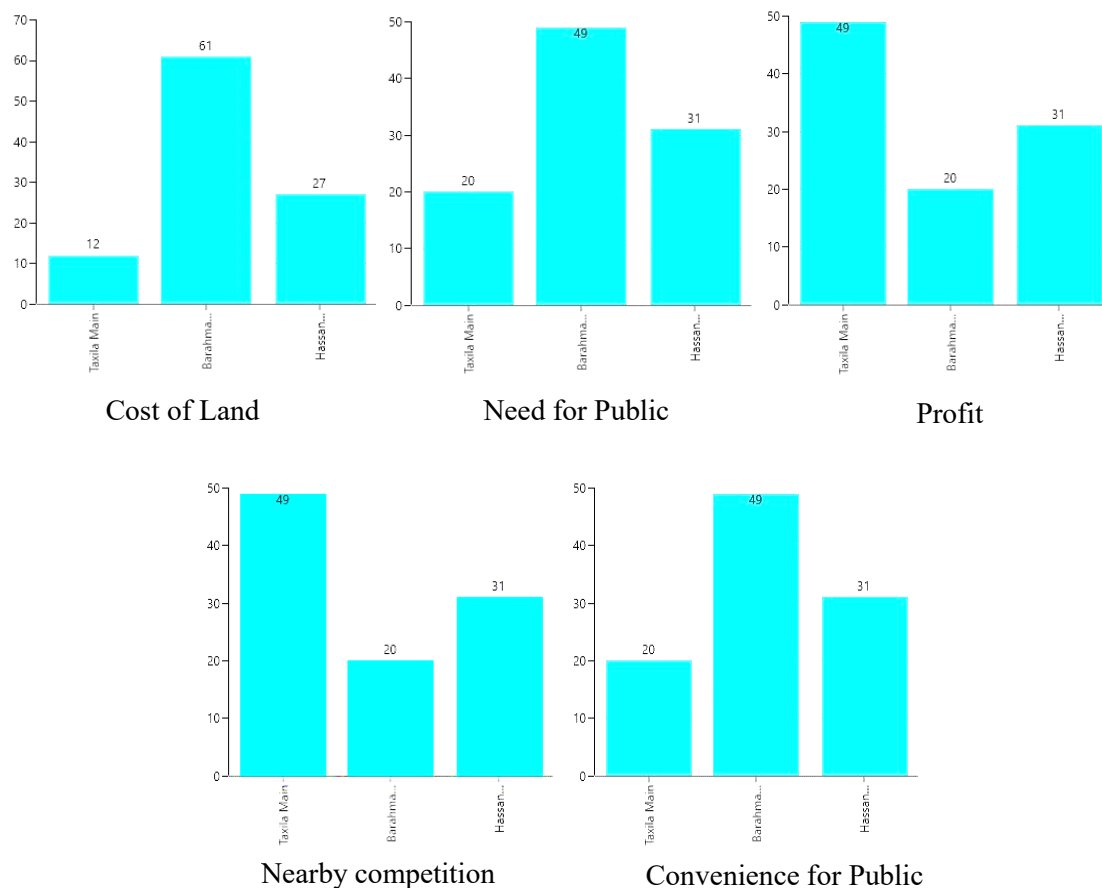


Figure 3.4 Weightage of location alternatives based upon five criteria's

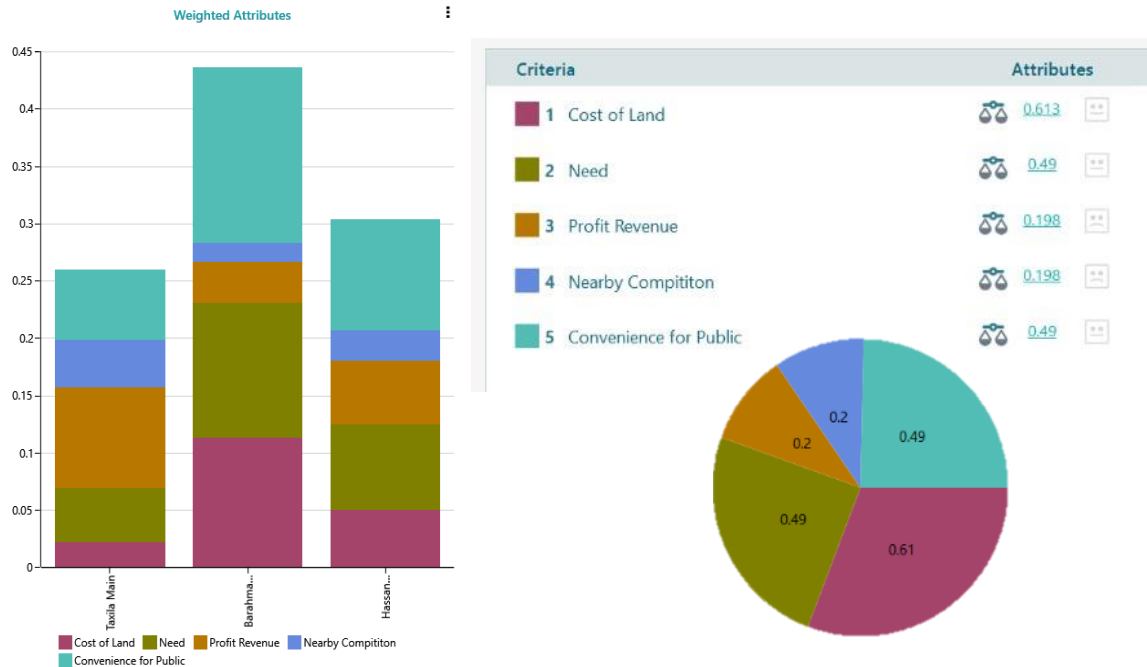


Figure 3.5 Final results of location alternatives

3.2. Layout planning

After successfully deciding the location, now it is to make a layout design [2] based on the available area. For creating the Layout, first a proposed layout is made under available space using Adobe software then this layout is optimized using a graph-based technique,

3.2.1. Proposed layout

The total area for facility is 900 X 500, the available working area is 500 X 200, the 200 from left and 200 from right is kept as Government area and 300 from front side on road can be used for parking purposes as shown in figure 3.6 of layout made on Adobe Illustrator

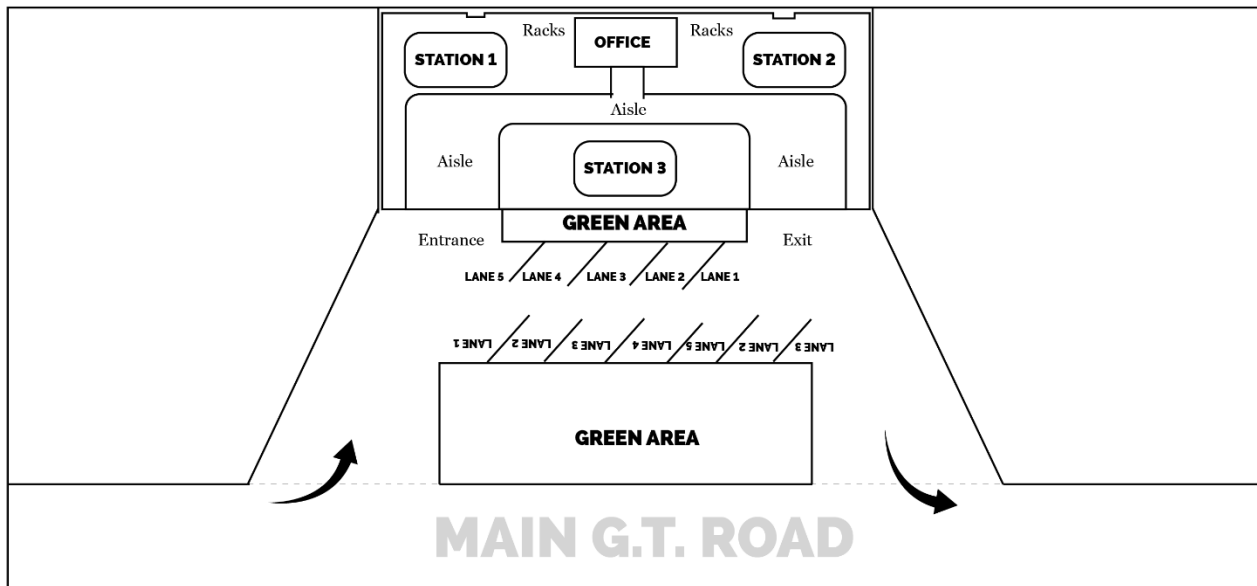


Figure 3.6 Layout of our proposed design

In this proposed design the no. of car workstations is 3, means three cars can be served at a time. There is an office kept in the left most side in the layout. There is an accessories area and an aisle for uninterrupted workflow.

3.2.2. Problems Identification

The problems with this layout are:

1. Three stations are almost in the center which may interrupt the workflow in the facility.
2. The cars may have difficulty in turning over and there is a great interruption in the IN and OUT flow for cars.
3. Aisle with is small, there is a small gap between the car stations

So there must be a layout design which maximize the flow of activities but the number of workstations should be kept same, there must be an office and material handling accessories area as well.

3.3. Optimizing the current Layout

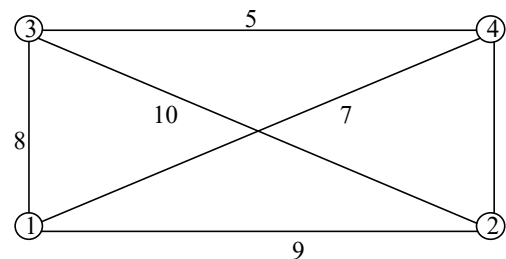
3.3.1. Graph based Method (Cognitive, PLO-2)

By using this technique, a rough data is assumed for the completing the requirements of calculations.

3.3.1.1. Graph Based Method for departments construction

From / To matrix for the required departments

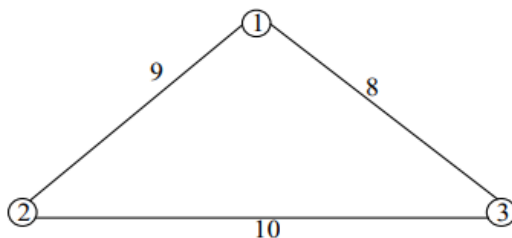
(Required data is assumed)



From/To	1	2	3	4
1	1	9	8	7
2		2	10	6
3			3	5
4				4

	2	3	
1	9	8	= 17
4	6	5	= 11

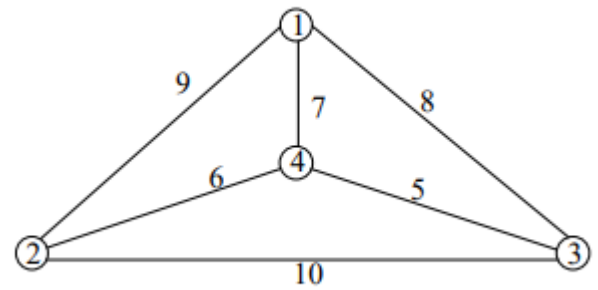
Larger →



Calculating the total distance

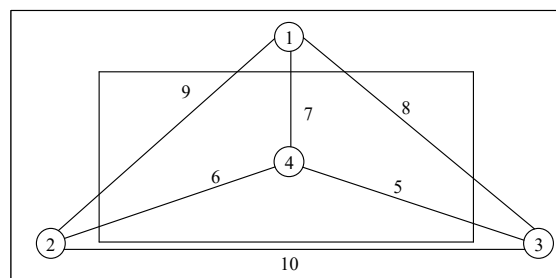
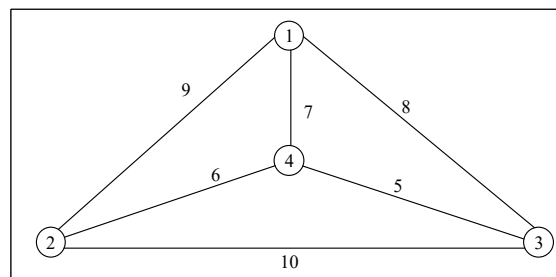
$$\begin{array}{c}
 | \quad 1 \quad | \quad 2 \quad | \quad 3 \quad | \\
 \hline
 4 \quad 4 \quad 6 \quad 5 \quad = \quad 15
 \end{array}$$

Faces	Total	
1-2-3	17	<u>Larger</u>
1-2-4	16	
1-3-4	15	

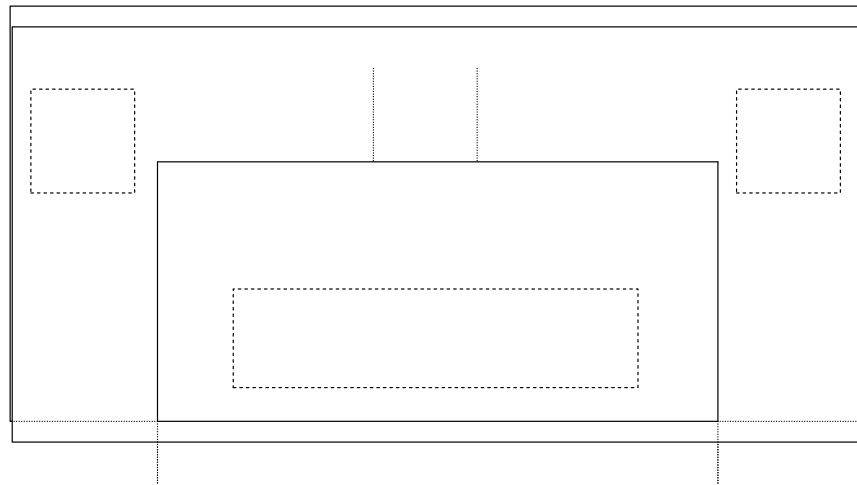


Arc	1-2	1-3	1-4	2-3	2-4	3-4	Total
Weight	9	8	10	7	6	5	45

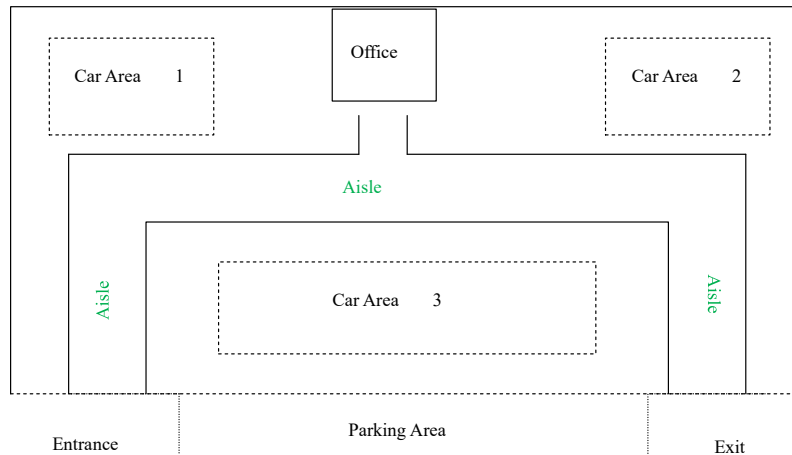
Construct the departments



Further modifications in the constructed departments



Label the departments with their respective names



3.4. The optimized layout

The layout design as shown below in figure, has maximum flow inside the stations, the work area seems spacious and there is an aisle which is open for all car stations, the IN and OUT flow is perfect without any interruption. This optimized layout covers all the requirements that must a facility have.

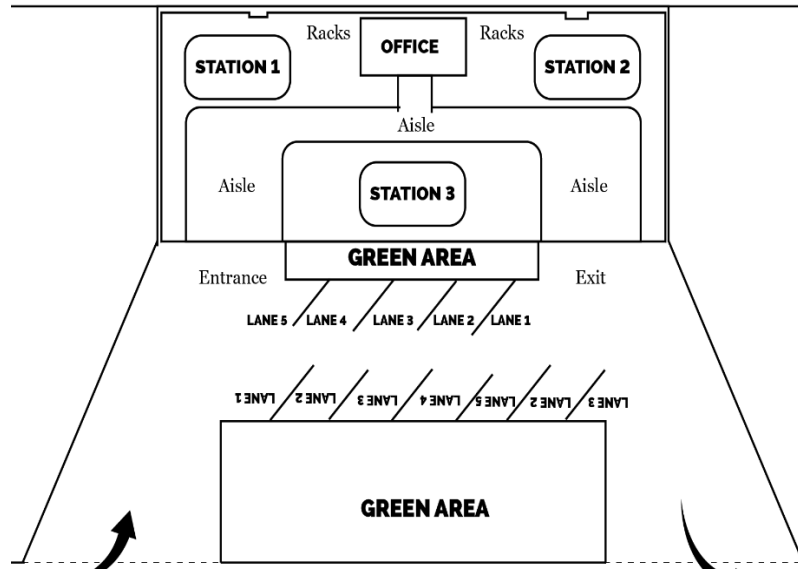


Figure 3.7 Layout of optimized design

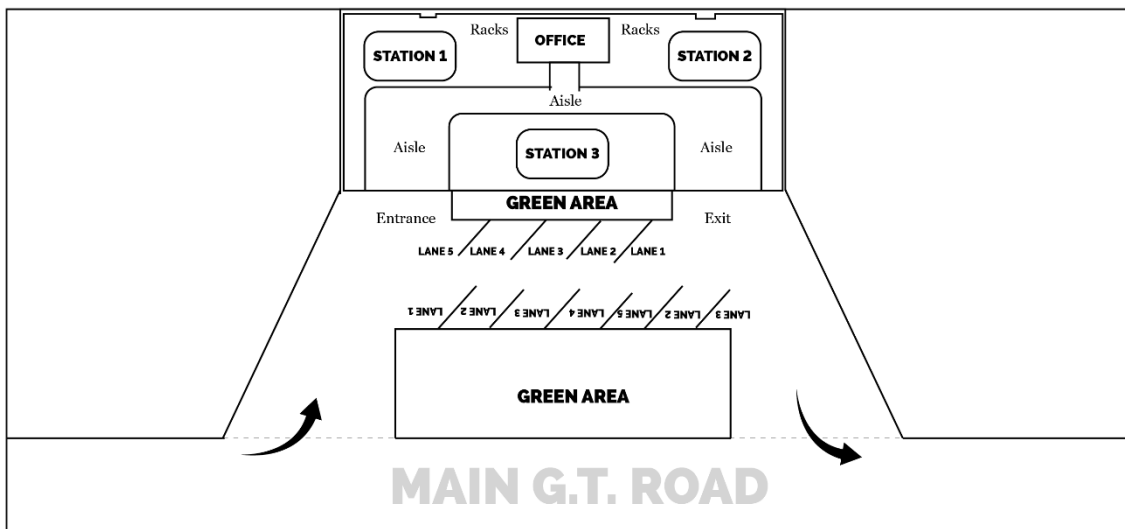


Figure 3.8 Complete Layout for Facility

3.5. Design of Facility (Cognitive, Affective, PLO-1, PLO-3, PLO-11)

The 2D design of the optimized layout is made in Adobe, which includes all the necessary equipment that a facility must have. Then the facility layout is optimized by using graphic based method.

3.6. Improvements and Strategic Planned Design (Cognitive, PLO-2)

3.6.1. Interior of Facility

The following are some important factors which contributes great in the efficient working environment:

- Lighting and Luminosity
- Fire Extinguisher and Emergency exit
- Material Handling Equipment and Office
- Over Head Crane
- Other machinery and tools of automobile Maintenance

3.6.1.1. Lighting and Luminosity

THE recommended light level for the normal working environment is around 30 foot-candles (average) or 300 lux (average). To maintain light in the shop the lights are placed on the roof. To spread the light throughout the workshop, white paint is established on all the walls and a four-foot yellow border beneath the walls.

3.6.1.2. Fire Extinguisher and Emergency exit (Cognitive, PLO-6)

According to OSHA standards there must be a fire extinguisher [14] provided within 50 feet of the working area. For this very reason, we have placed 2 fire extinguishers at the entrance and the exit side in the shop.



Figure 3.9 Fire extinguisher and emergency exit in a facility

OSHA requires at least two exit routes [15] to allow quick evacuations of employees and other building occupants during an emergency. In our facility plan there are two evacuations first one is the entrance and second one is the exit gate. In case of emergency any of the side can be used for evacuation. There is also an emergency exit in our design.

3.6.1.3. Material Handling Equipment and Office

There are cabinets attached with the walls, for accessories keeping and best material handling. There is a 100 X 50 size office, which is mandatory for dealing with clients, keeping record and other management related stuff.

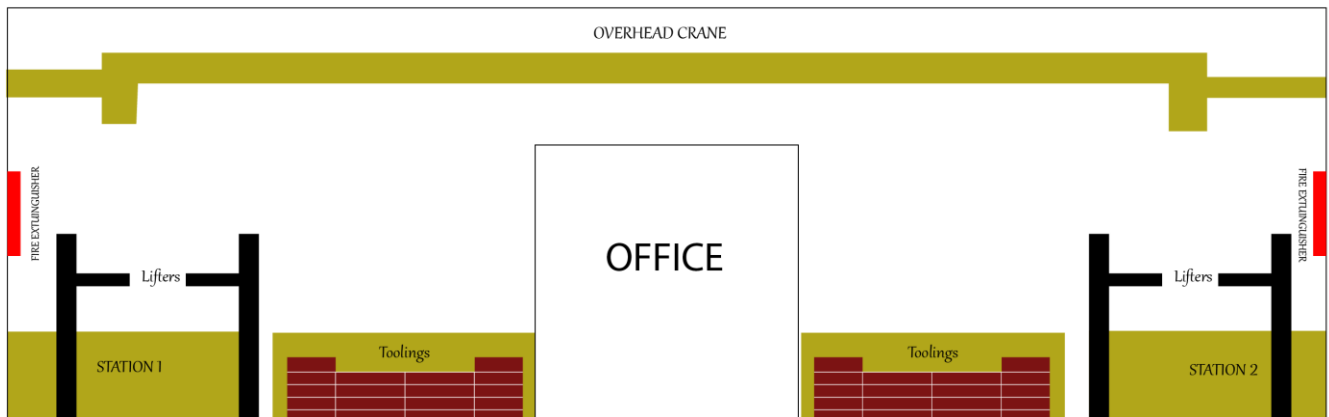


Figure 3.10 Front view of crane and tooling

3.6.1.4. Over Head Crane

Overhead Cranes are used in the sop to support processing and handling throughout the workshop, the dead cars can easily be lifted and transported to their respective workstation by using this crane, placed with roof of facility as shown in figure.

3.6.1.5. Other machinery and tools of automobile Maintenance

The other tools which an auto workshop must have are Car lifting Jack, tyres carrying racks, oil drums, tyres jacks, air spray gun, air compressor etc. Below figures show the necessary equipment that we have put in the facility.

3.7. Exterior design

3.7.1. Car Parking

The parking area important for those clients who are with their multiple cars and for those who are in waiting for their turn, so the parking area is designed to maximize the no, of cars in available space. There are three lanes each with 45^0 angle maintained in the front of facility as shown below in figure 3.11.

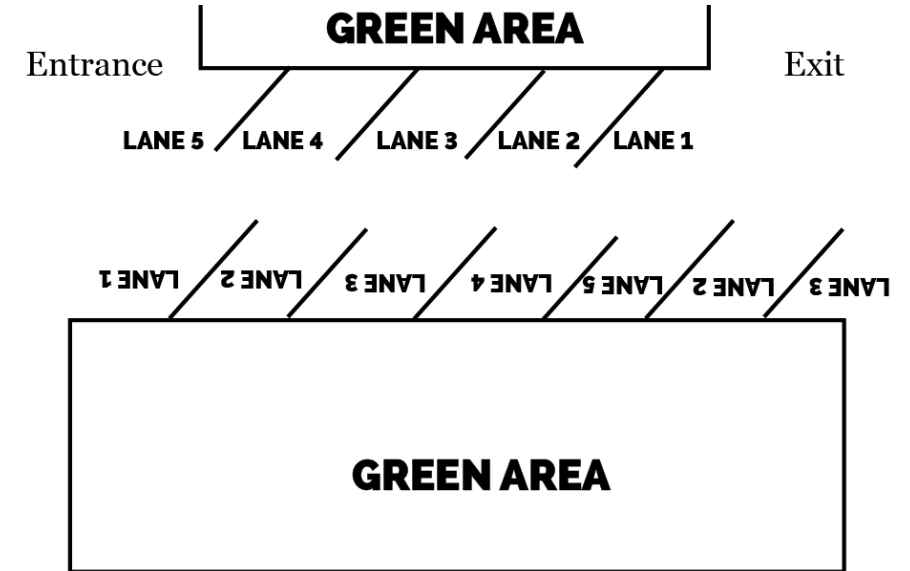


Figure 3.11 Parking lines are at an angle 450

3.7.2 Advertisement Board and Streetlights

An advertisement and symbolic board is displayed prior 500 meter from the facility, so that the customer must become aware about the workshop before it passed away. Also for lighting, the light pole is displayed in the center which lit the facility at night time.

3.7.3 Greenbelt for Environment friendly system

Greenbelts [16] are effective tools for mitigation of traffic smoke and noise pollution. A green belt is placed in the front of facility at the available non useable area as shown below in figure 3.12.

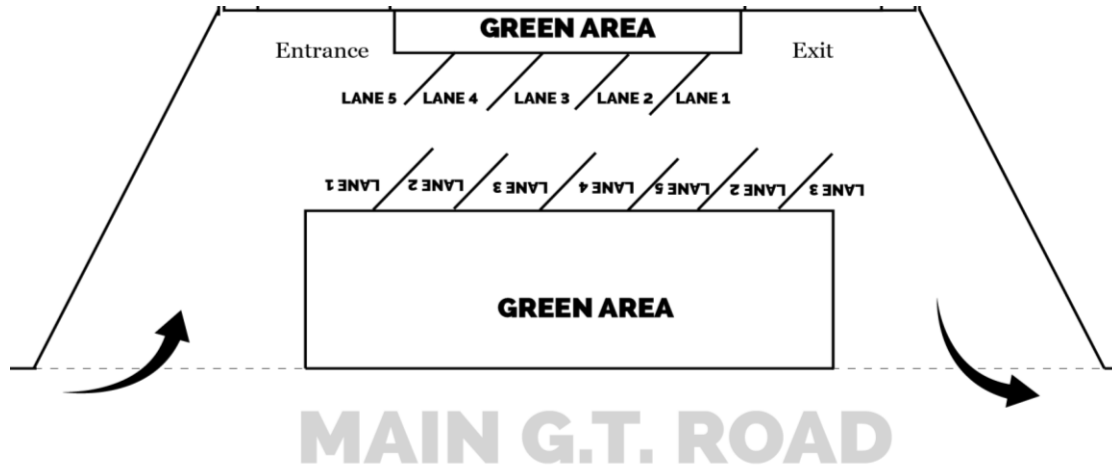


Figure 3.12 Green area is kept in front of facility

Conclusion

Facility design is an important component of strategic planning, for it provides an accurate assessment of the institution's physical environment and its working. The auto workshop near on the road side provides people a way to get their cars repaired quickly and effectively. The Importance of Auto Repair and workshops include maintenance which prevents further breakdowns to the vehicle, It Saves Money and time. So it is concluded that, the facility is designed after considering all important factors of facility planning and optimization techniques which were the basic requirements of the project.

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